

12. Ecosystem

- **Ecosystem** is the interaction of living things among themselves and with their surrounding environment.
- There are two basic ecosystems – terrestrial and aquatic.

Structure of Ecosystem

- The interactions between the various biotic and abiotic factors of an ecosystem lead to the maintenance of the ecosystem.
- Stratification is the vertical distribution of the different species occupying the different levels. For example, trees occur at a higher level than shrubs.
- The various aspects taken into consideration to study the functioning of ecosystem are:
 - Productivity
 - Decomposition
 - Energy flow
 - Nutrient cycling

Productivity

- A constant supply of sunlight is required for the proper functioning of any ecosystem.
- The amount of biomass produced per unit area over a time period by plants during photosynthesis is defined as the **primary productivity**.
- It is expressed as weight (g m^{-2}) or energy (Kcal m^{-2}).
- Primary productivity depends upon
 - Type of plant species inhabiting a particular area
 - Photosynthetic capacity of plants
 - Nutrient availability
- Productivity can be mainly divided into gross primary productivity (GPP) and net primary productivity (NPP). GPP is the rate of production of organic matter during photosynthesis.

$$\text{NPP} = \text{GPP} - \text{Respiratory losses (R)}$$

- Annual net productivity for whole biosphere is about 170 b tons of organic matter.
- **Secondary productivity** is defined as the rate of formation of new organic matter by consumers.

Decomposition - It is the process of breakdown of complex organic matter into inorganic substances such as carbon dioxide, water, nutrients, etc.

- **Fragmentation** – Breaking down of detritus (dead plant and animal remains, faecal matter) into smaller particles by detritivores (decomposers)
- **Leaching** Process by which these inorganic matters enter the soil
- **Catabolism** – Process by which detritus is degraded into simpler inorganic substances by bacterial and fungal enzymes.
- **Humification** – Accumulation of humus in the soil. Humus is resistant to microbial action and decomposes at an extremely slow rate. It acts as a reservoir of nutrients.

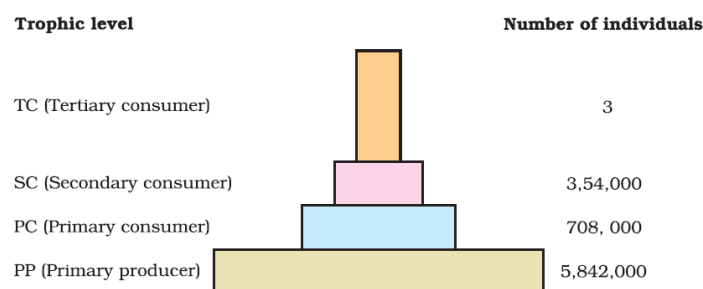
- **Mineralization** – Process by which humus further degrades to release minerals into the soil. It is an oxygen consuming process and is controlled by the chemical composition of detritus and climatic conditions.

Energy Flow

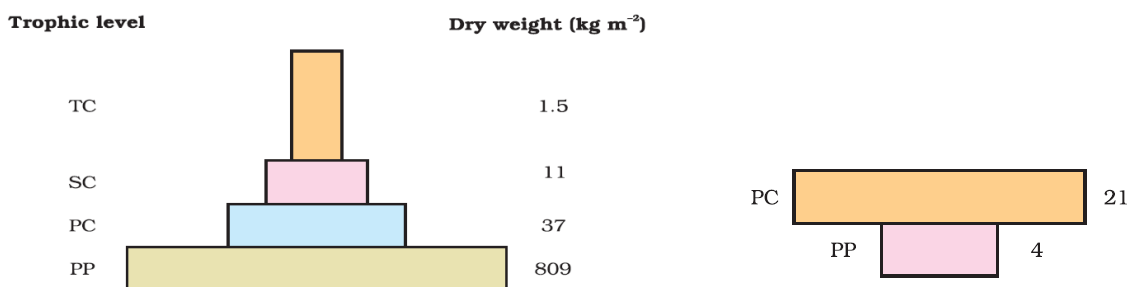
- Sun is the sole source of energy for all ecosystems on the earth.
- Plants and other photosynthetic organisms utilize less than 50% of the solar radiation known as the **photosynthetically active radiation (PAR)**.
- In an ecosystem, plants are called **producers** and all animals depend upon the plants directly or indirectly for their food. Hence, they are known as consumers or heterotrophs.
- The consumers can be further divided into primary consumers (herbivores), secondary consumers (primary carnivores), and tertiary consumers (secondary carnivores).
- **Food chain** – the energy flow among the various constituent animals is known as the food chain.
- **Food web** – the interconnection of the various food chains is called the food web.
- **Trophic level** – every organism occupies a specific level in their food chain known as the trophic level.
- **Standing crop** – each trophic level contains a certain amount of living material at a certain time known as the standing crop.
- The number of trophic levels in a food chain is restricted since the energy transfer follows the **10 percent law** i.e., only 10% of the energy is transferred from a lower trophic level to a higher one.

Ecological Pyramids

- The energy relationship between the different trophic levels is represented by the ecological pyramids.
- Their base represents the producers or the first trophic level while the apex represents the tertiary or top level consumer.
- Ecological pyramids are of 3 types:
 - (a) Pyramid of number
 - (b) Pyramid of biomass
 - (c) Pyramid of energy

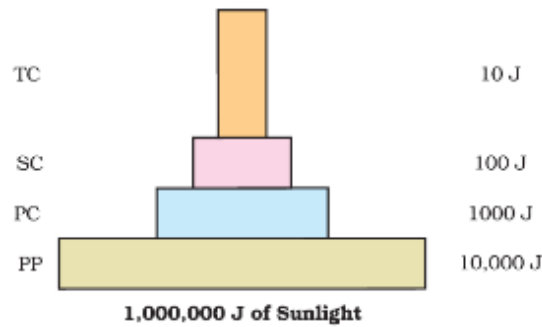


(a) Pyramid of numbers in a grassland ecosystem. Only three top-carnivore are supported in an ecosystem based on production of nearly 6 million plants



(b) Pyramid of biomass shows a sharp decrease in biomass at higher trophic levels

Inverted pyramid of biomass-small standing crop of phytoplankton supports large standing crop of zooplankton



(d) An ideal pyramid of energy. Observe that primary producers convert only 1% of the energy in the sunlight available to them into NPP

- In most ecosystems, the three pyramids are upright except in some cases:
 - The pyramid of biomass is inverted in an ocean ecosystem since a small standing crop of phytoplankton supports a large number of zooplankton.
 - The pyramid of number can be inverted when, say, a large tree is eaten by small insects.
 - However, the pyramid of energy is always upright.
- A trophic level represents a functional level and not a single species as such. Also, a single species may become a part of more than one trophic level in the same ecosystem at the same time depending upon the role it plays in the ecosystem.
- **Limitations of ecological pyramids:**
 - The ecological pyramids do not take into account the same species belonging to more than one trophic level.
 - It assumes a simple food chain that almost never exists in nature. It does not explain food webs.
 - Saprophytes are not given a place in ecological pyramids even though they play a vital role in ecosystem.